

CENG381 Stochastic Processes

Finite Markov Chains - Question Set 1

Q1 (30 points). Consider the Markov chain on states $\{0, 1, 2\}$ with transition matrix:

	0	1	2
State 0	[? 0.4 0.2]		
State 1	[0.3 ? 0.3]		
State 2	[0.1 0.3 ?]		

- Replace each '?' with the correct value so that P is a valid stochastic matrix.
 - Compute the two-step transition matrix P^2 .
 - Find $p_{02}^{(3)}$, the probability of moving from state 0 to state 2 in exactly three steps.
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Q2 (35 points). A frog lives on two lily pads: East (E) and West (W). Each day it jumps to the other pad with probability p (from E) or probability q (from W), and stays with the complementary probability.

- Write the 2×2 transition matrix P in the order (E, W).
 - Starting from E, what is the probability of being at E after exactly 2 jumps? Express your answer in terms of p and q .
 - Compute P^2 symbolically and identify the pattern in terms of p and q . What does this suggest about the long-run behaviour?
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Q3 (35 points). A small company tracks employee satisfaction each week as one of three states: Satisfied (S), Neutral (N), or Dissatisfied (D). The weekly transition probabilities are:

From S: stay S with prob 0.7, go to N with prob 0.2, go to D with prob 0.1

From N: go to S with prob 0.3, stay N with prob 0.5, go to D with prob 0.2

From D: go to S with prob 0.1, go to N with prob 0.4, stay D with prob 0.5

- Write the 3×3 transition matrix P (rows and columns in order S, N, D).
- Starting from state N, compute the probability distribution over $\{S, N, D\}$ after 2 weeks by evaluating $\mu_1 = (0, 1, 0) * P$ and $\mu_2 = \mu_1 * P$.
- Apply the Chapman-Kolmogorov equation $p_{ij}^{(n+m)} = \sum_k p_{ik}^{(n)} * p_{kj}^{(m)}$ to find $p_{SD}^{(3)}$ using your P^2 from part (b) and the original P .